Community energy: close the gap with local operations

Brief for Colombian utilities and foundations: public energy-poverty data, the real cost of rural OPEX, and cooperatives as a sustainability vehicle.

1. The national gap

15.4%
IMPE incidence 2024 (~8.1 M people in energy poverty)[1]

9.5%
2030 scenario target (~5.17 M people; −2.9 M vs 2024)[1]

75.92% / 93.12%
Rural vs national residential electric coverage (ICEE 2024)[2]

63.1% / 52.8%
Departmental IMPE La Guajira / Chocó (persons, 2024)[1]

At the observed pace (~300,000 people per year leaving energy poverty), lifting ~2.9 M people takes ~9.7 years from 2024 (single-digit incidence ~2034).[1] The IMPE 2030 scenario needs ~490,000 people per year × 6 years (≥1.0 pp/year incidence).[1]

Pace ~300k people/year
2.9 M ÷ 300k ≈ 9.7 years
~2034

Scenario ~490k people/year
490k × 6 ≈ 2.94 M
2030

2. Reframe: OPEX, not CAPEX alone

- **Contraloría:** sustainability concern where methodologies to pay for operation, microgrid administration, and off-grid PV are missing; maintenance and monitoring failures.[3]
- **Amazonas:** ~COP 30 billion (royalties) solar investment not operating; damaged batteries — "critical state" on sustainability.[4]
- **CREG:** thousands of rural PV systems without management → "often abandoned"; missing standards, certified installers, and O&M.[5]

[1] Multidimensional Energy Poverty Index (IMPE), 2025 report, 3rd edition (2024 base year), pp. 12, 22, 33.

[2] UPME, ICEE Bulletin 2024; UPME press release on 93.12% coverage (2024).

[3] El Colombiano, Contraloría on solar projects in remote areas.

[4] Seguimiento.co / Procuraduría, abandoned Amazonas project.

[5] CREG GEN-Renewables report (Circular 005-13); CREG Concept 4022/2019 (O&M).

3. The cooperative as the tool

CTH PROPOSAL

Community O&M + revenue + local ownership (energy cooperative) complements CAPEX with explicit OPEX contracts (PV, batteries, monitoring; CREG/SSPD) and community governance; IMPE access/quality KPIs — does not replace ICEE/IPEM.

4. Local talent — Escuela Garavito

Escuela Colombiana de Ingeniería Julio Garavito — local talent and green jobs: CLSA chair and energy-sector challenges; 3rd place Solucionatón 10X (2026); renewable diploma/courses and 101.76 kWp lab.[6,7]

5. Territorial partner PASO Colombia

Productive and cooperative projects in high energy-poverty departments — **PASO does not run energy projects.**[10]

- **Fonseca, La Guajira** — COOMPAZCOL, ASMUPERIJA (IMPE 63.1%)
- **Puerto Asís / Puerto Guzmán, Putumayo** (IMPE 58.7%)
- **Istmina, Chocó** — Bioconversión Chocó, Aserrío Continental (52.8%)
- **San José del Guaviare** (IMPE 38.9%)
- **Arauca** — COOPCACAO, COMPROCAR, Filimarpaz (27.9%)

Energy Coop candidate community: Corozal, Sucre (IMPE 43.1%) — not a PASO project.[10]

6. Data lab — filtering method

Funnel from ~1,000 MinEnergía prioritised communities to final candidates; filters: solar resource, ZNI/off-grid, legal ID (NIT), social licence (scores not shown).[10]



7. Partner network (target, no signed agreements)

FMO and **RVO** (Colombia solar finance/facilitation).[8,9] **Green Choice** and **Energie Samen** — finance and peer learning for community energy.[10]

[6,7] escuelaing.edu.co — CLSA, Solucionatón 10X, continuing education, Renewable Energy Laboratory.

[8] fmo.nl — Colombia projects. [9] Gutami MOU 2023; commercial facilitation in the solar ecosystem.

[10] CTH / PASO, 2026 (project brief and GRP application filter).

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